Data Engineering & Analytics For Yelp Restaurant Data

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Executive summary

Objective

Build an analytics-ready data model for Yelp to generate business insights.

Approach

- Transformed raw Yelp data into a snowflake schema (OLAP-ready)
- Applied data cleaning and standardization
- Designed fact and dimension tables

Result

- Faster, simpler analytical queries
- Supports slicing by business, location, category, and ratings
- Interactive Tableau dashboards

Business Case

Yelp’s raw review data is semi-structured and not optimized for analytical querying.

Business requires a structured, analytics-ready data model to support reliable reporting and insights.

Data Source Overview

Yelp Open Dataset (U.S. Metropolitan)

Source	Kaggle （around 2010-2019）
Files	business_data; reviews_data; users_data
Format	Json
Key Fields	business_id, user_id, review_id, city, stars, categories
Geographic Coverage	Covers 15 U.S. states (metro areas)
Scope	Contains restaurants, supermarkets, services

U.S. Census Population Data

Source	data.census.gov
Files	census_data (by place in each States)
Format	Excel
Key Fields	city, total population (2010 state population)
Geographic Coverage	Covers 15 U.S. states, 600+ cities

Data Cleaning and Processing

Business Data → dim_restaurant

Format: Json

Shape: 150 thousand lines

Category: groceries, services, etc.

Type: multiple attributes for one business

City: one city several name

1. Drop non-restaurant business
2. Organize 13 restaurant type (fast food, italian, ect.)
3. Unified City Naming

Format: CSV

Shape: 30 thousand lines

Category: Restaurant only

Type: 1 attribute for one restaurant

City: one city one name

Data Cleaning and Processing

Review Data

Format: Json

Shape: 8 million lines

1. Extract 200 thousand lines
2. Match by business id in restaurant file
3. Normalized review date

dim_review

Format: CSV

Shaple: 100 thousand lines

User Data

Format: Json

Shape: 2 million lines

1. Extract 200 thousand lines
2. Match by User id in dim_review file

dim_user

Format: CSV

Shaple: 200 thousand lines

Missing Data

Restaurant Filtering

- Categories are null-filled, converted to lowercase, and filtered by the keyword *"restaurant"*
- Records without a detected cuisine type are removed

Dimension Table Processing

- dim_category: remove null values before generating category list
- dim_user: missing or blank elite values treated as 0 years
- dim_date: generated from cleaned review timestamps

Text Standardization

- Restaurant names and addresses are normalized
- Missing values are preserved (no imputation applied)

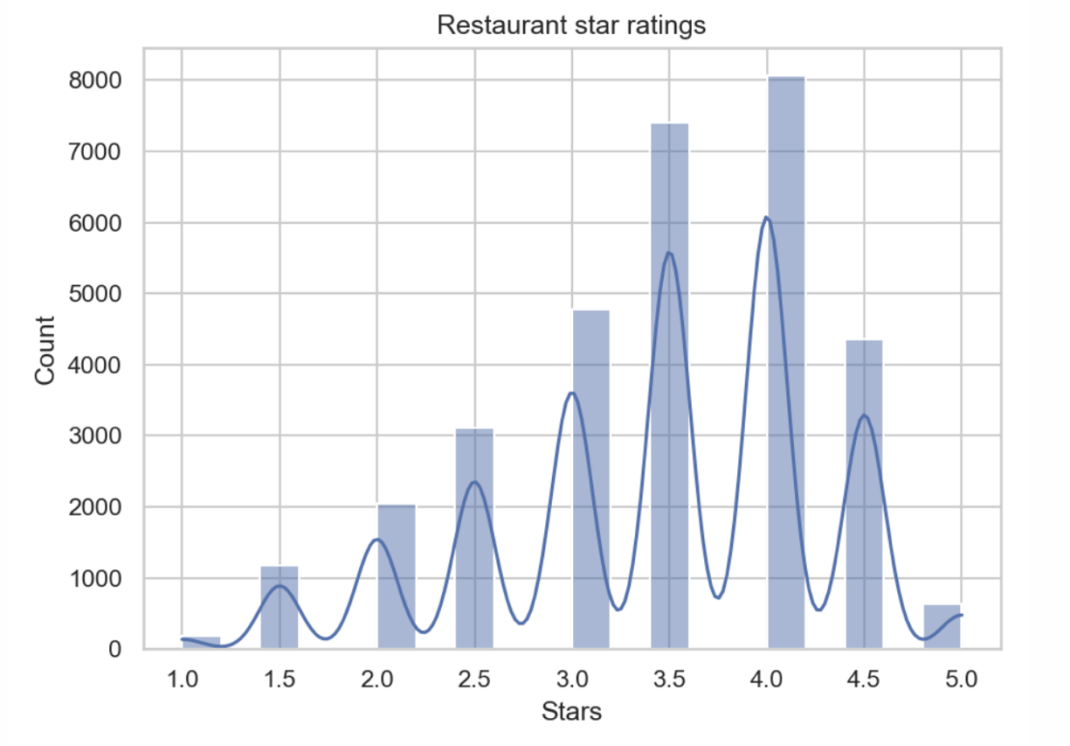
Fact Table Validation

- Reviews kept only if business_id exists in dim_restaurant

City Key Validation

- After merging city_key, NULL values are counted and reported
- No artificial filling is applied to missing city references

EDA

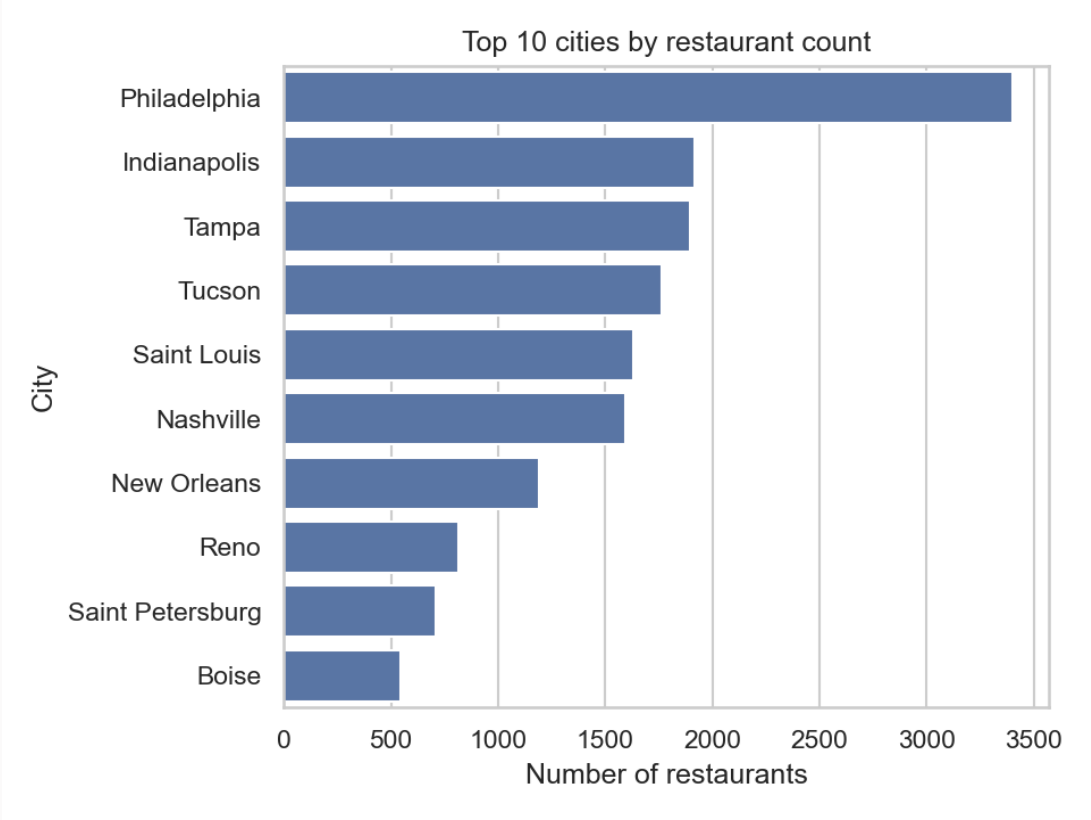


The main peaks are in the range of 3.0, 3.5, 4.0, and 4.5, indicating that most restaurants are concentrated in the mid-to-high score range, especially in the area of 3.5–4.0.

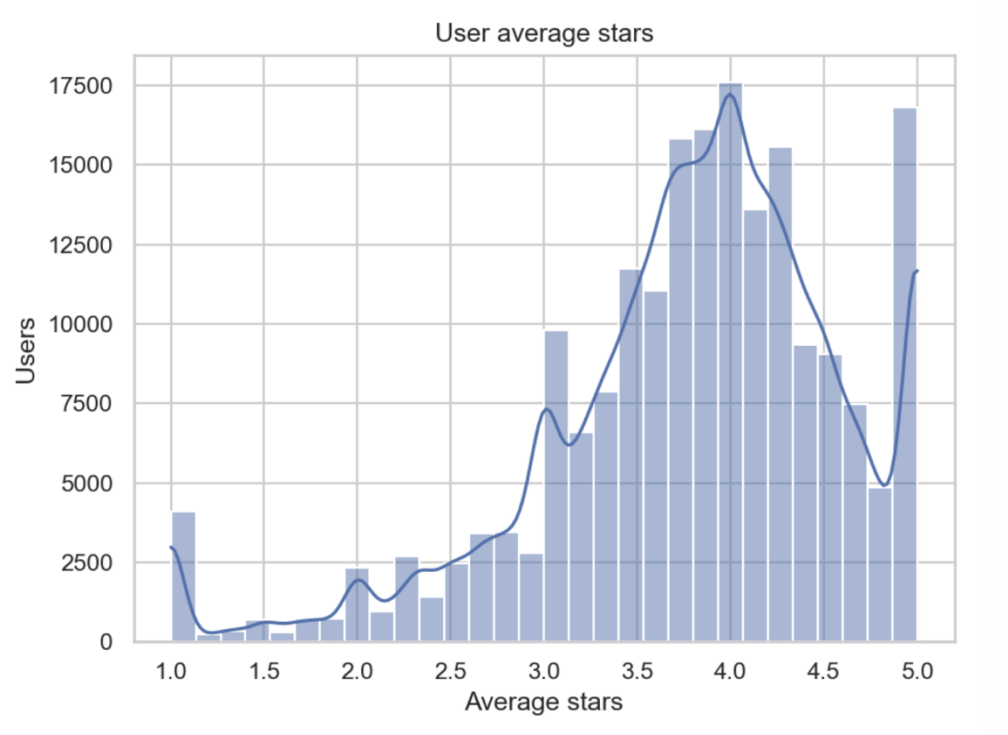
EDA

Horizontal axis: Number of restaurants;
Vertical axis: City name.

Restaurant distribution is highly concentrated in a few large cities, the cities on the list also represent a more competitive market.



EDA



Horizontal axis: Users' average star ratings (1–5).

Vertical axis: Number of users in each rating range.

Most users fall between 3.7–4.2, showing generally high ratings. The mean is about 3.77.

The 5-star peak is likely from users with very few reviews and all ratings of 5..

The left side (≤2.5) is very small, indicating fewer strict or low-rating users.

Methodology & Tools

Methodology

ETL Pipeline

Built analytical visualizations using Tableau

Tools

- Python
- Excel
- Google Cloud
- MySQL
- MySQL Workbench
- Tableau

Data Model Overview

A dimensional analytical model using snowflake schema

Key entities:

- Restaurant
- Review
- Restaurant Category
- User
- Location

High-level relationships:

- Restaurant has many Reviews
- User writes Reviews
- Restaurant located in City / State / Region
- Restaurant classified into Category

Model Structure & Tables

Fact table

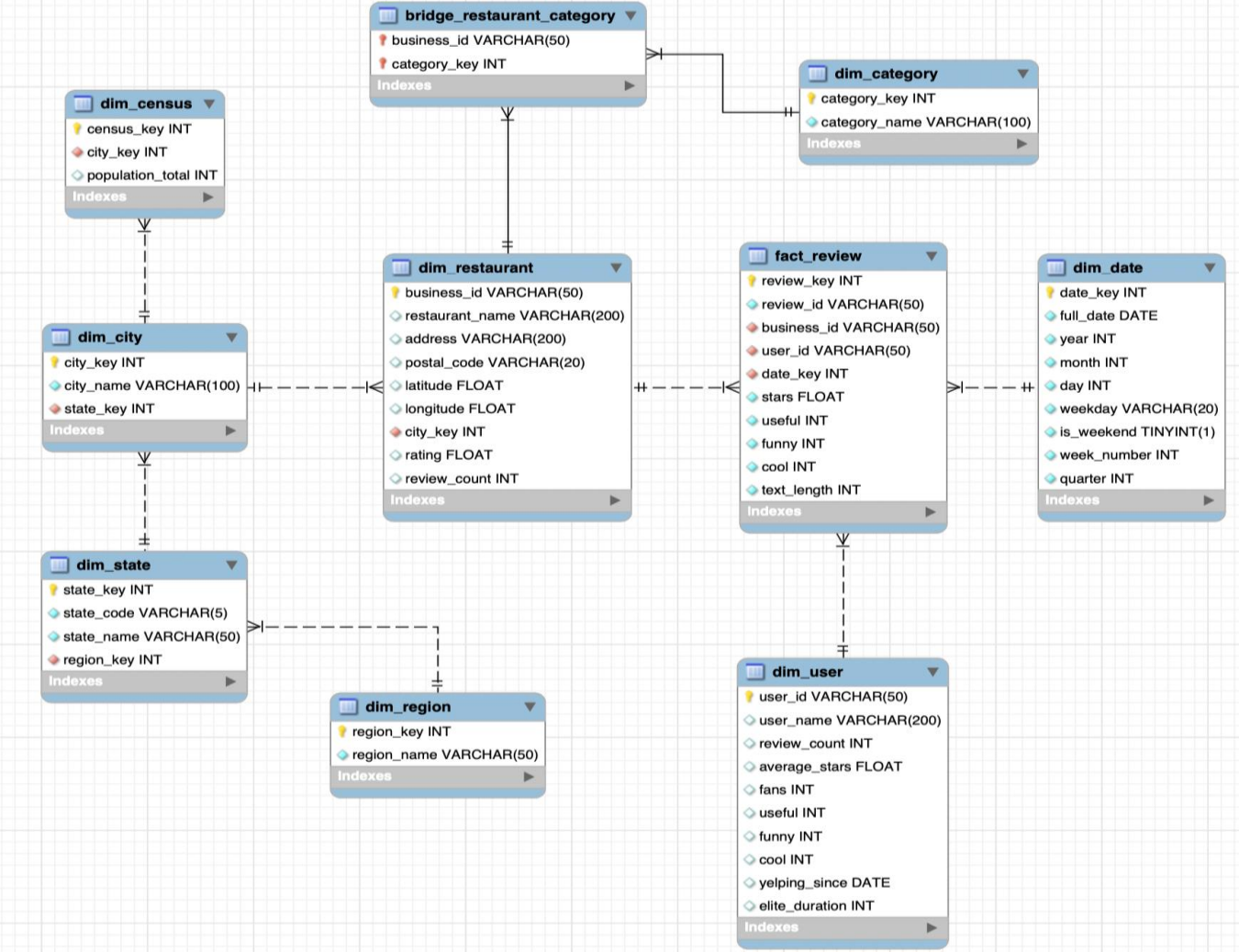
`fact_review`

Dimension tables

- `dim_restaurant`, `dim_category`
- `dim_user`, `dim_date`
- `dim_city`, `dim_state`, `dim_region`
- `dim_census`

Bridge table

`bridge_restaurant_category`


bridge_restaurant_category
business_id VARCHAR(50)
category_key INT
Indexes
dim_category
category_key INT
category_name VARCHAR(100)
Indexes
dim_census
census_key INT
city_key INT
population_total INT
Indexes
dim_restaurant
business_id VARCHAR(50)
restaurant_name VARCHAR(200)
address VARCHAR(200)
postal_code VARCHAR(20)
latitude FLOAT
longitude FLOAT
city_key INT
rating FLOAT
review_count INT
Indexes
fact_review
review_key INT
review_id VARCHAR(50)
business_id VARCHAR(50)
user_id VARCHAR(50)
date_key INT
stars FLOAT
useful INT
funny INT
cool INT
text_length INT
Indexes
dim_date
date_key INT
full_date DATE
year INT
month INT
day INT
weekday VARCHAR(20)
is_weekend TINYINT(1)
week_number INT
quarter INT
Indexes
dim_city
city_key INT
city_name VARCHAR(100)
state_key INT
Indexes
dim_state
state_key INT
state_code VARCHAR(5)
state_name VARCHAR(50)
region_key INT
Indexes
dim_region
region_key INT
region_name VARCHAR(50)
Indexes
dim_user
user_id VARCHAR(50)
user_name VARCHAR(200)
review_count INT
average_stars FLOAT
fans INT
useful INT
funny INT
cool INT
yelping_since DATE
elite_duration INT
Indexes

Insights

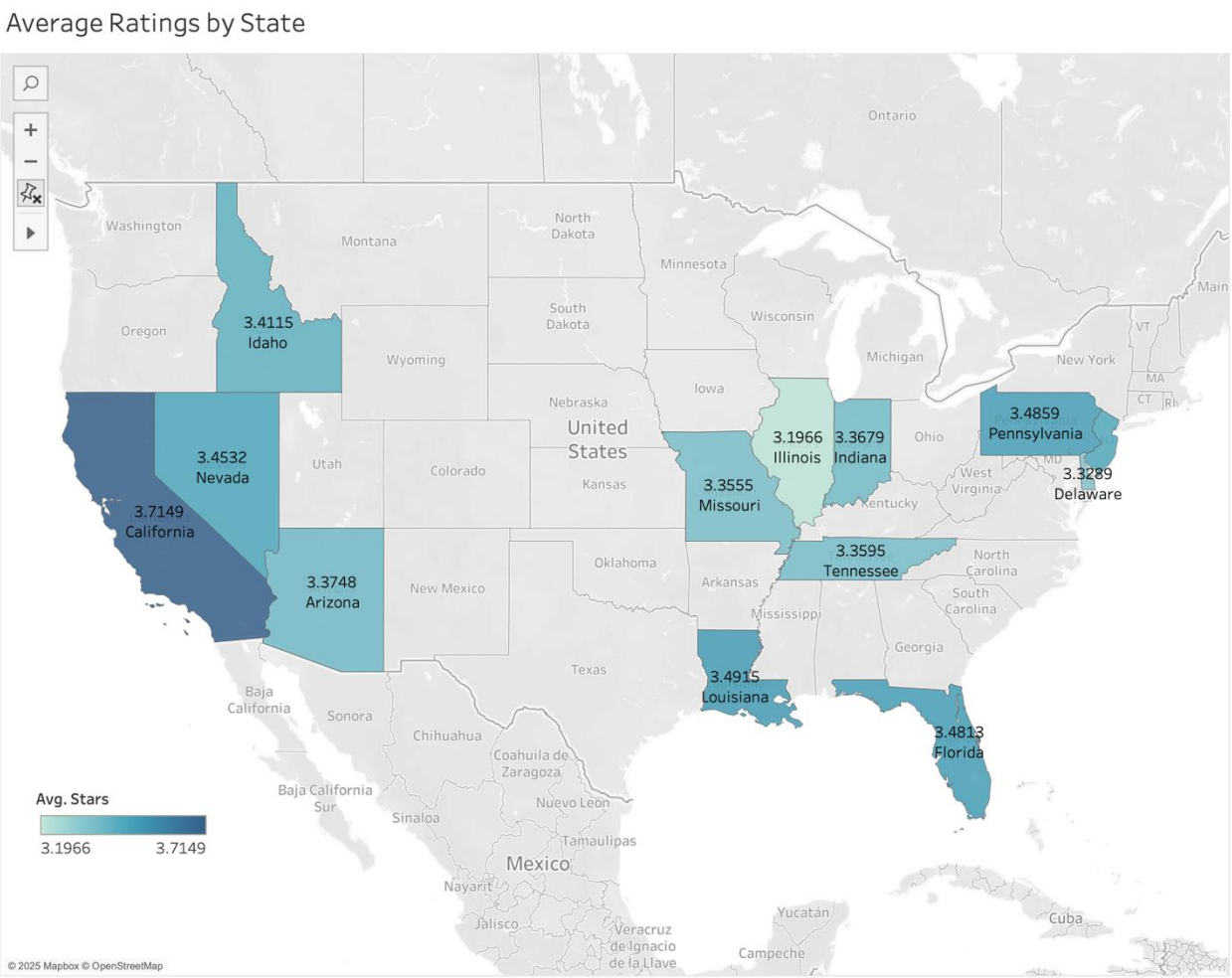
Average Ratings by State

California and Pennsylvania are the top-rated states in this dataset

Central states show lower ratings overall

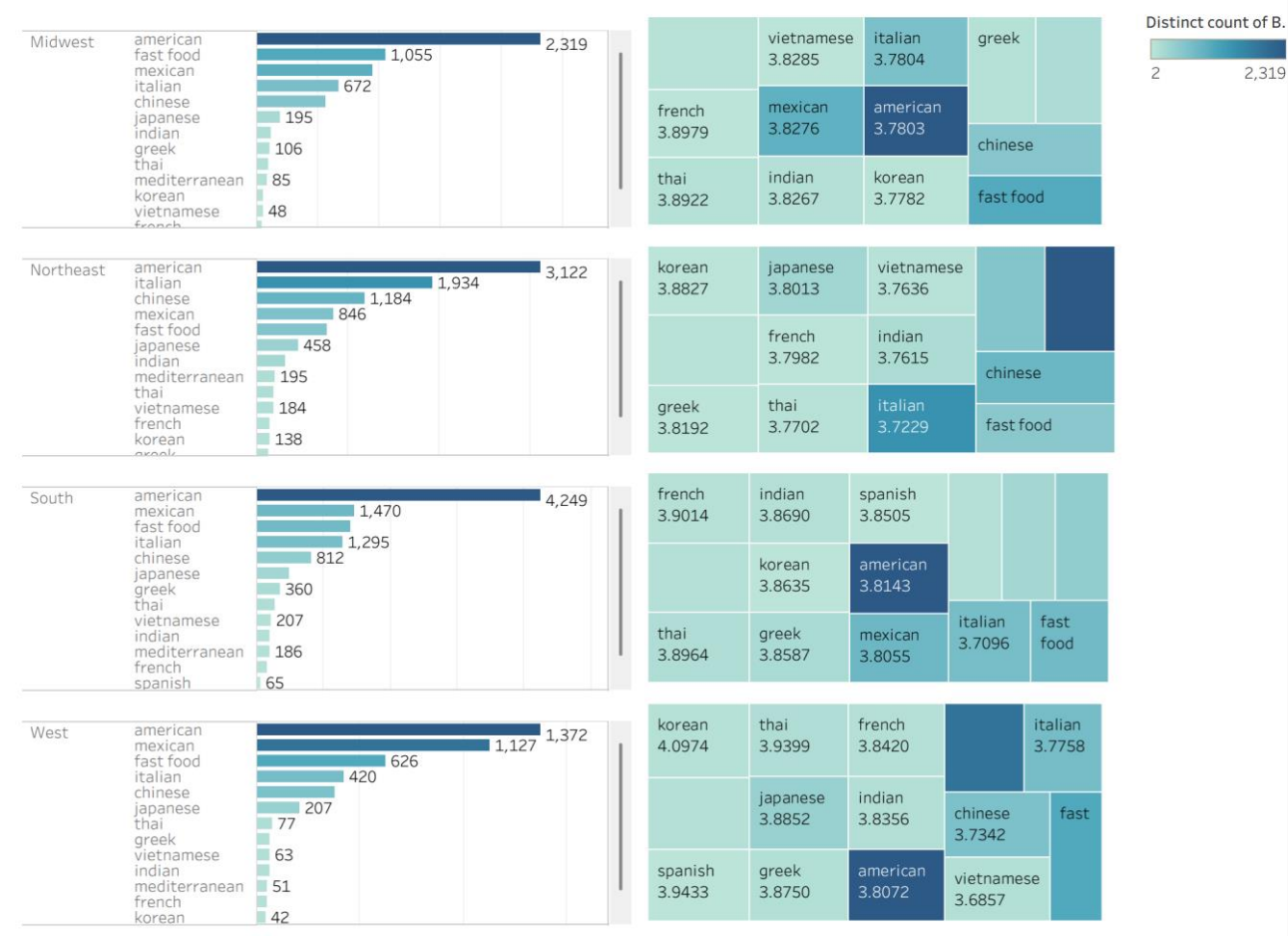
Midwest and central states (e.g., Illinois, Indiana) tend to have lower average ratings.

Recommendation: Popular States are more saturated, good for benchmarking but not idea for expansion.



Insights

Restaurant Categories Across U.S. Regions: Distribution and Performance



Insights

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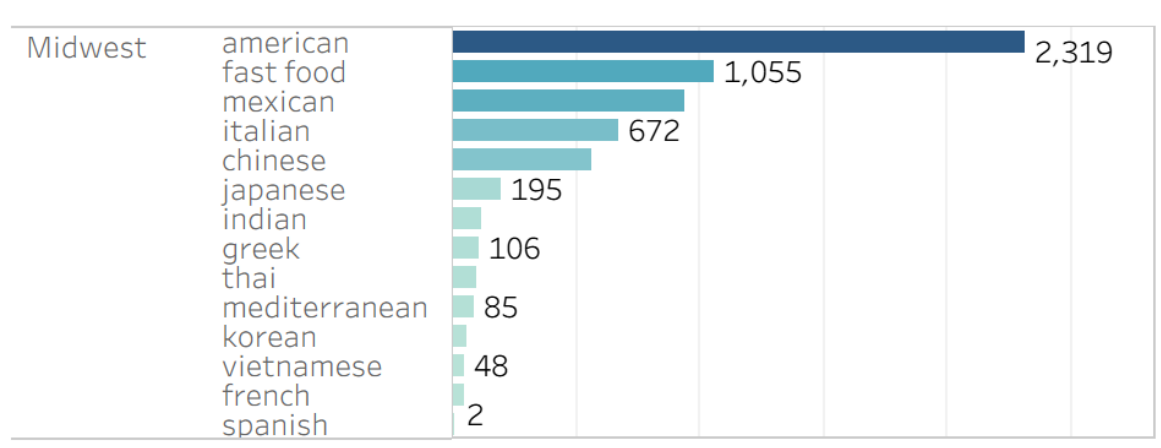
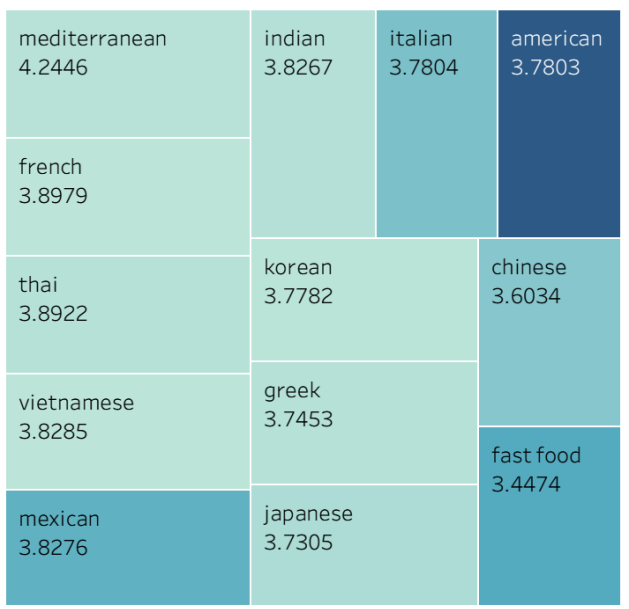
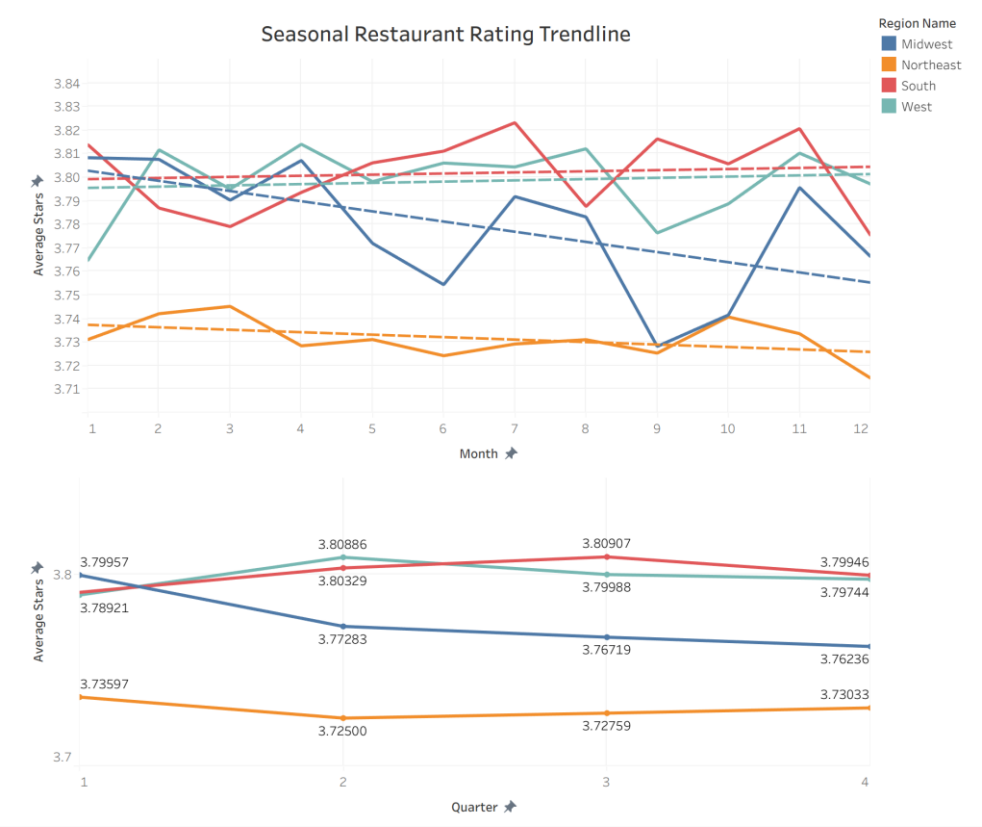




- American cuisine dominates across all regions
- Fast Food tend to have lower average ratings compared to other cuisines.
- Regional rating patterns are stable (Average ratings for the same cuisine type are relatively consistent across regions, implying that cuisine type influences ratings more than location.)

Recommendation: Introduce higher rated cuisines such as Mediterranean, French, or Thai in low rated areas.

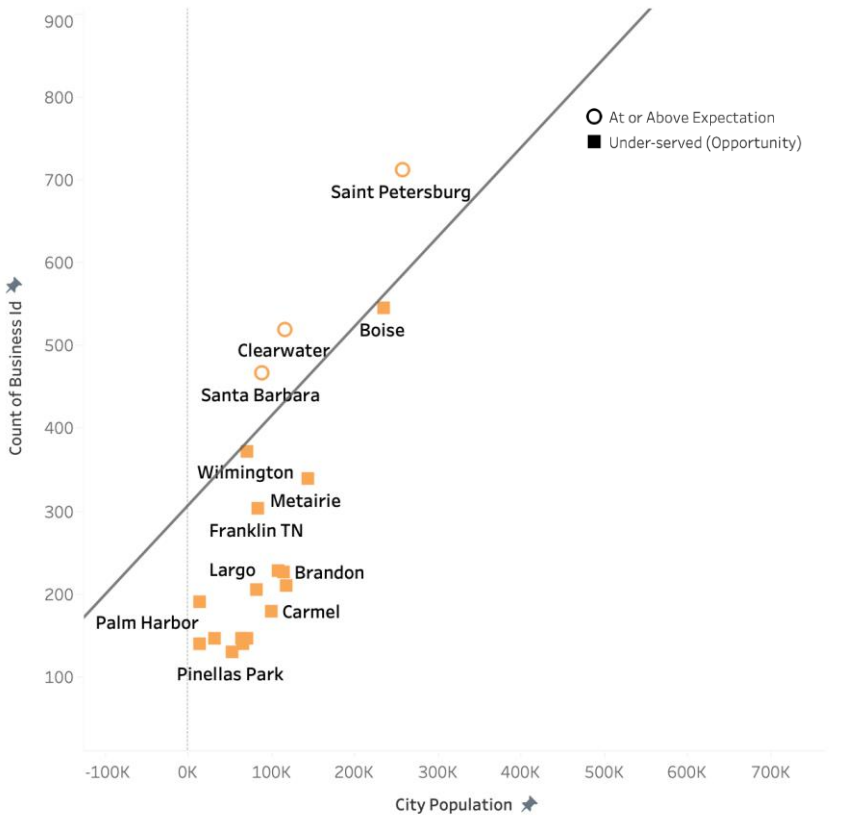
Insights Seasonal Restaurant Rating Trendline



- **Regional differences outweigh seasonal effects**
  South and West consistently have higher average ratings
- **Midwest shows the most volatility**
  Noticeable rating drops occur in late summer and early winter.

Recommendation: Region with lower ratings reinforce services quality in specific month; Yelp prompt localized support programs

Insights Restaurant Count vs. City Population



- The **diagonal trendline** represents the *expected number of restaurants* based on city population

- Cities above the trendline (e.g., *Cleawater*) have more restaurants than expected → highly competitive / saturated markets.

Recommendation: Cities below the trendline (e.g., *Brandon*) have fewer restaurants than expected → underserved markets with strong growth opportunities

Recommendation & Lessons learned

1

- Expand into underserved markets
- Prioritize cuisine types with strong, consistent ratings
- Enhance Yelp's data model to support operational insights
- Strengthen location-level analysis with additional demographic data

2

- Semi-structured data requires heavy transformation
- Consistent keys are essential for integrating large datasets
- Dimensional modeling greatly improves analytical capabilities
- EDA reveals insights only after data quality issues are resolved

Thank You!

Q&A

Any questions?

Reference

Yelp dataset: https://www.kaggle.com/datasets/yelp-dataset/yelp-dataset/data

Census Data:
https://data.census.gov/table?q=race